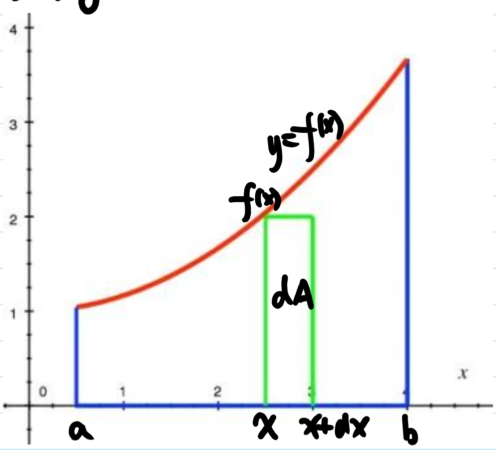
6.2定积分在几何上的应用

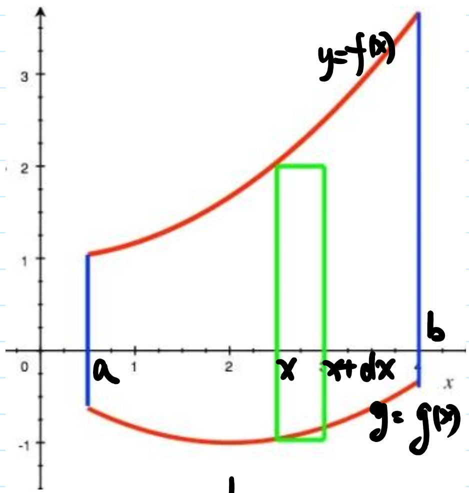
2017年12月18日 8:10

§2. 定积分在几何中的应用.

1. 平面图形的面积

(1) 直角坐标系

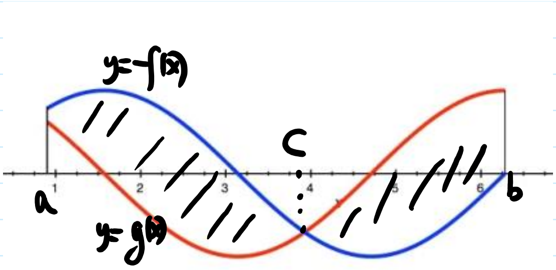




$$A=\int_a^b f(x)\,dx$$

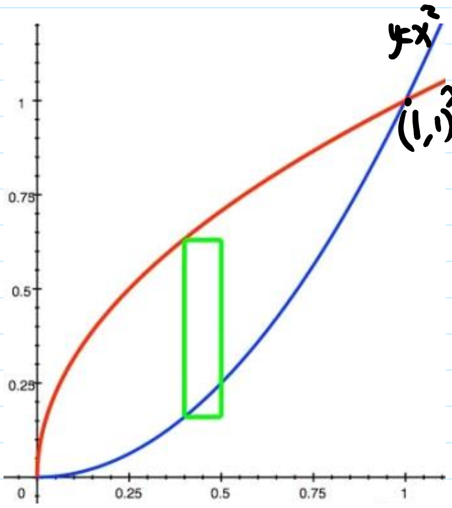
$f(x)dx$ — 面积元素.

$$A=\int_a^b [f(x)-g(x)]\,dx$$

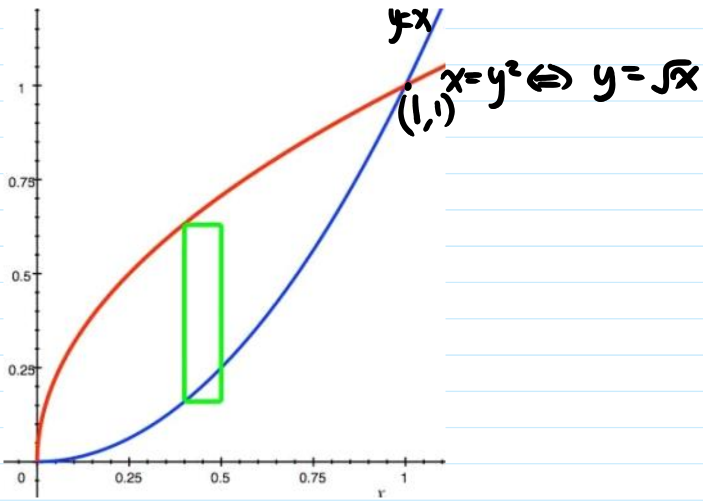


$$A=\int_a^b |f(x)-g(x)|\,dx=\int_a^c [f(x)-g(x)]\,dx+\int_c^b [g(x)-f(x)]\,dx$$

例1. 求由 $y=x^2$ 和 $x=y^2$ 所围成的平面图形的面积.

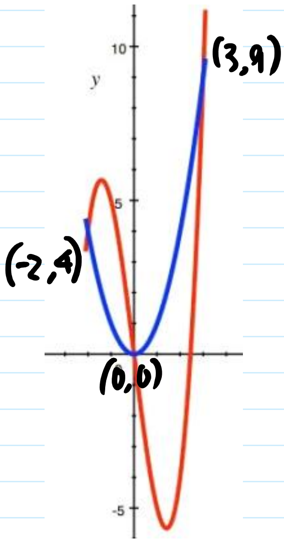


$x=y^2 \Leftrightarrow y=\sqrt{x}$



解: $A = \int_0^1 (\sqrt{x} - x^2)\,dx = \frac{1}{3}$

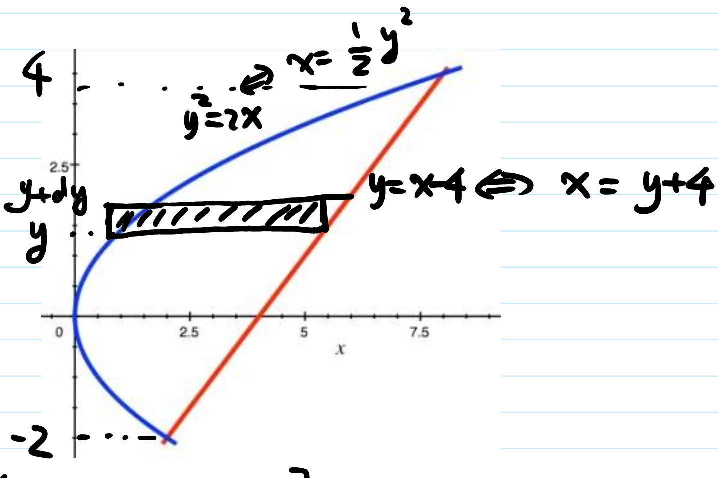
例2. $y = x^3 - 6x$. $y = x^2$



解: $A = \frac{253}{12}$

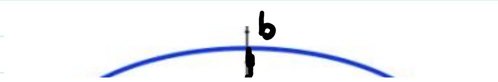
$= \int_{-2}^{3} \left| (x^3 - 6x) - x^2 \right| dx$

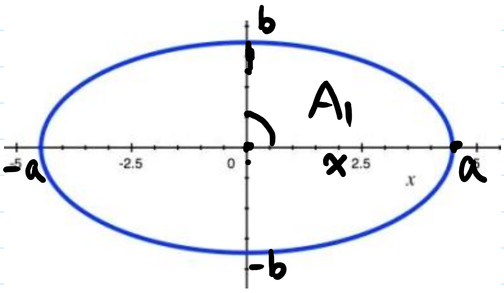
例3. $y^2 = 2x$. $y = x - 4$.



$A = \int_{-2}^{4} \left[(y+4) - \frac{1}{2} y^2 \right] dy = 18$.

例4. 求椭圆 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ 的面积





解：由对称性，

$$A = 4A_1 = 4\int_0^a f(x)\,dx = 4\int_0^a y\,dx$$

$$\begin{cases} x = a\cos t \\ y = b\sin t \end{cases}$$

$$A = 4\int_{\frac{\pi}{2}}^{0} b\sin t(-a\sin t)\,dt = 4ab\int_0^{\frac{\pi}{2}} \sin^2 t\,dt$$

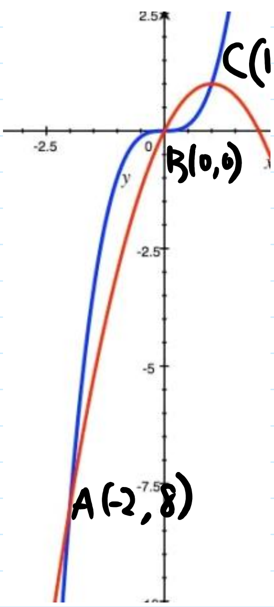
$$= 4ab\cdot\frac{1}{2}\cdot\frac{\pi}{2} = \pi ab.$$

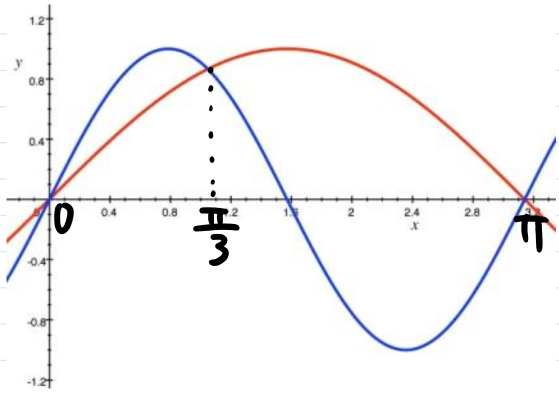
注： $L:\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$ $\quad t:\alpha\to\beta \iff x: a\to b$

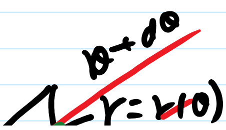
$$A = \int_a^b y\,dx = \int_\alpha^\beta \psi(t)\varphi'(t)\,dt.$$

例：(1) $y = x^3$, $y = 2x - x^2$

(2) $y = \sin x$, $y = \sin 2x$ $\quad [0, \pi]$



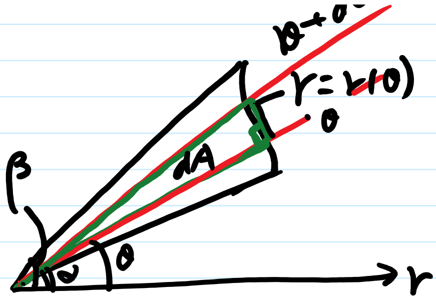




(2) 极坐标系

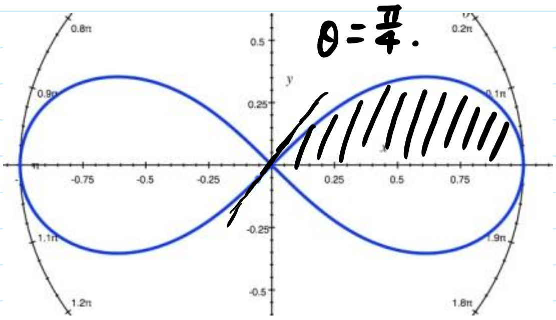
$\theta: \alpha \to \beta$.

$r = r(\theta)$



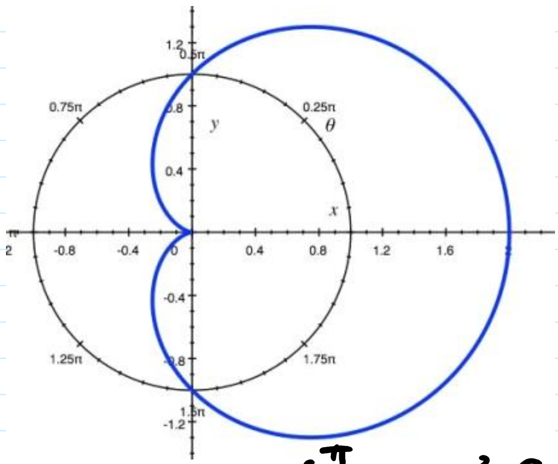
$$A = \int_\alpha^\beta dA = \frac{1}{2}\int_\alpha^\beta r^2(\theta)\, d\theta$$

例5. 求曲线 $\rho^2 = a^2\cos 2\theta$ 所围图形的面积



解: $A = 4A_1 = 4\int_0^{\frac{\pi}{4}} \frac{1}{2} a^2\cos 2\theta\, d\theta = a^2$.

例6. 求心形线 $r = a(1+\cos\theta)$ 所围图形的面积



解. $A = 2A_1 = 2\int_0^\pi \frac{1}{2} a^2(1+\cos\theta)^2\, d\theta = \frac{3}{2}\pi a^2$

练习: (3) 求两圆 $r = \sqrt{3}a$ 与 $r = 2a\cos\theta$ 公共区域的面积.

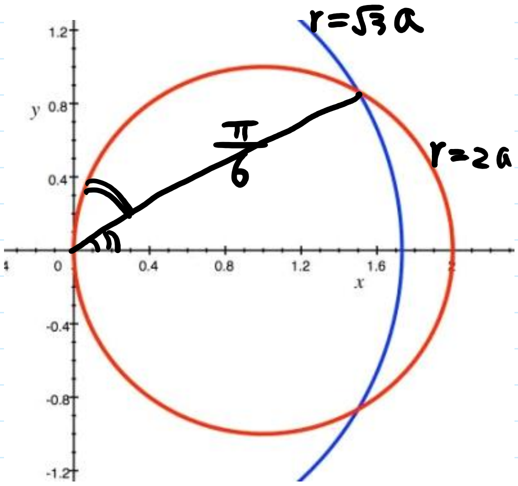
注: 圆的极坐标方程: $x = r\cos\theta$, $y = r\sin\theta$

(1) $x^2+y^2=R^2$; $r=R$, $0\le\theta\le 2\pi$.

(2) $(x-a)^2+y^2=R^2$: $r=2R\cos\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$.

(3) $x^2+(y-a)^2=R^2$: $r=2R\sin\theta$, $0\le\theta\le\pi$.

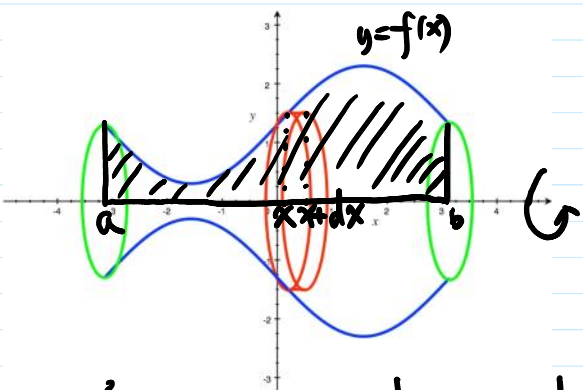
解：



$$A=\frac{7}{6}\pi a^2-\frac{\sqrt{3}}{2}a^2$$

2. 体积

(1) 旋转体的体积（平面图形绕直线旋转而成的立体）

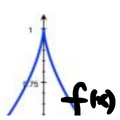


$dV=\pi f^2(x)\,dx$ —— 体积元素

$$V=\int_a^b dV=\int_a^b \pi f^2(x)\,dx$$

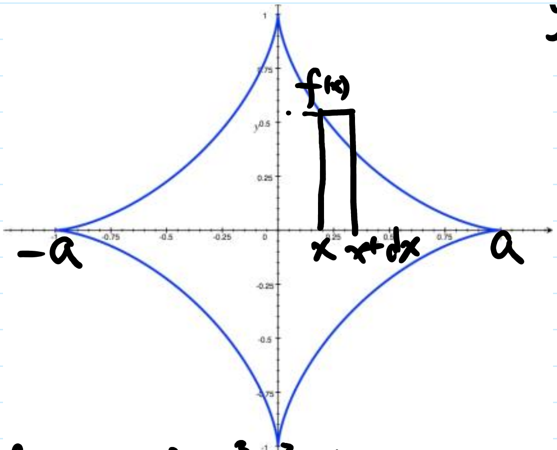
例7. 求星形线 $x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}$ $(a>0)$ 绕 x 轴旋转一周构成的旋转体的体积.

解：



$$y^2=\left(a^{\frac{2}{3}}-x^{\frac{2}{3}}\right)^3$$

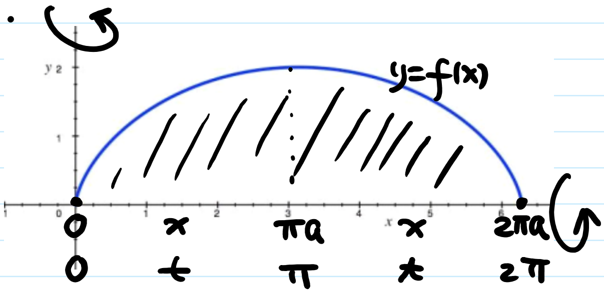
解:



$$V=2\int_0^a \pi\left(a^{\frac{2}{3}}-x^{\frac{2}{3}}\right)^3 dx=\frac{32}{105}\pi a^3.$$

例8. 求摆线 $\begin{cases} x=a(t-\sin t) \\ y=a(1-\cos t) \end{cases}$ 的一拱 $(0\le t\le 2\pi)$

与 x 轴所围图形分别绕 x 轴、y 轴所得旋转体的体积.



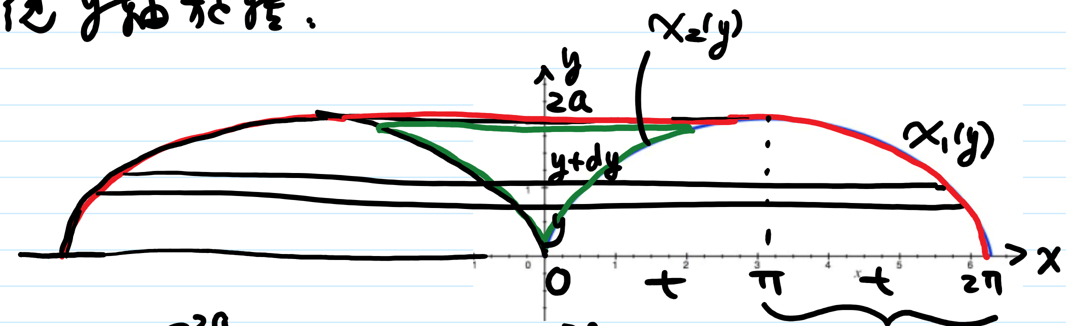
解: (1) 绕 x 轴旋转:

$$V_x=\int_0^{2\pi a}\pi f^2(x)\,dx=\pi\int_0^{2\pi}a^3(1-\cos t)^3\,dt$$

$$=\pi a^3\int_0^{2\pi}\left(1-3\cos t+3\cos^2 t-\cos^3 t\right)dt$$

$$=5\pi^2 a^3$$

(2) 绕 y 轴旋转.



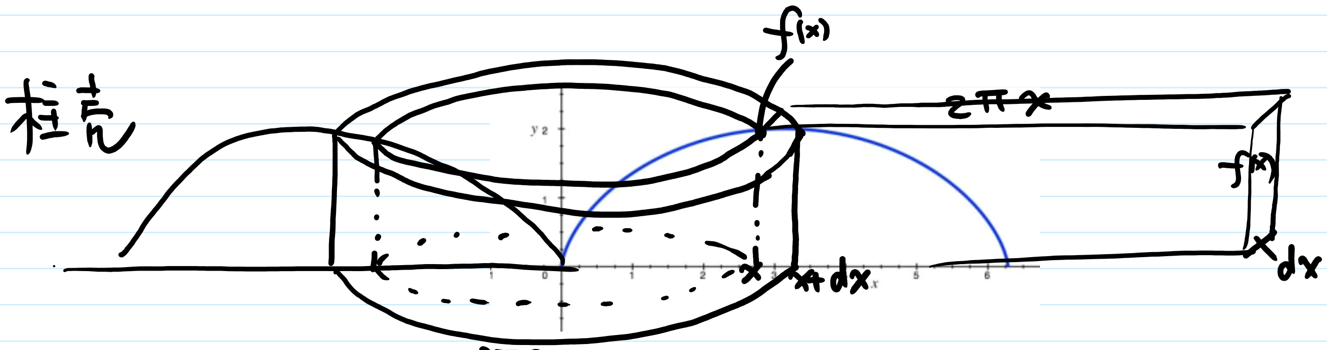
$$V_y=\int_0^{2a}\pi x_1^2(y)\,dy-\int_0^{2a}\pi x_2^2(y)\,dy$$

$$V_y = \int_0 \pi x_1^2(y)\,dy - \pi x_2^2(y)\,dy$$

$$= \pi\int_{2\pi}^{\pi} a^2(t-\sin t)^2\, a\sin t\, dt - \pi\int_0^{\pi} a^3 (t-\sin t)^2 \sin t\, dt$$

$$= 6\pi^3 a^3$$

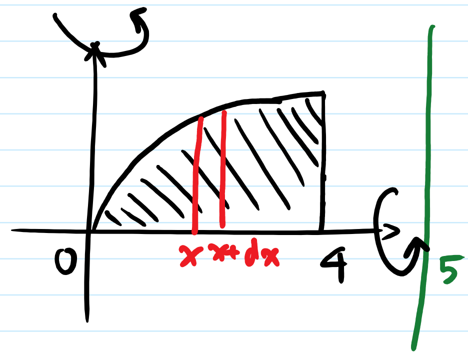
柱壳



$$V_y = \int_0^{2\pi a} \underline{2\pi x}\ \underline{f(x)}\ \underline{dx}$$

$$= 2\pi\int_0^{2\pi} a(t-\sin t)\, a^2(1-\cos t)^2\, dt$$

例19. $y=\sqrt{x}$. $x=0$. $x=4$ 及 x轴围成一平面区域 求该区域分别绕x轴. y轴旋转所成的旋转体的体积.

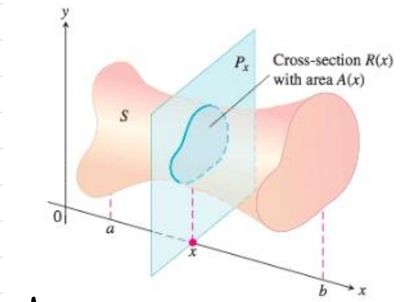


解: $V_x = \int_0^4 \pi f^2(x)\,dx$

$$= \int_0^4 \pi x\,dx = 8\pi.$$

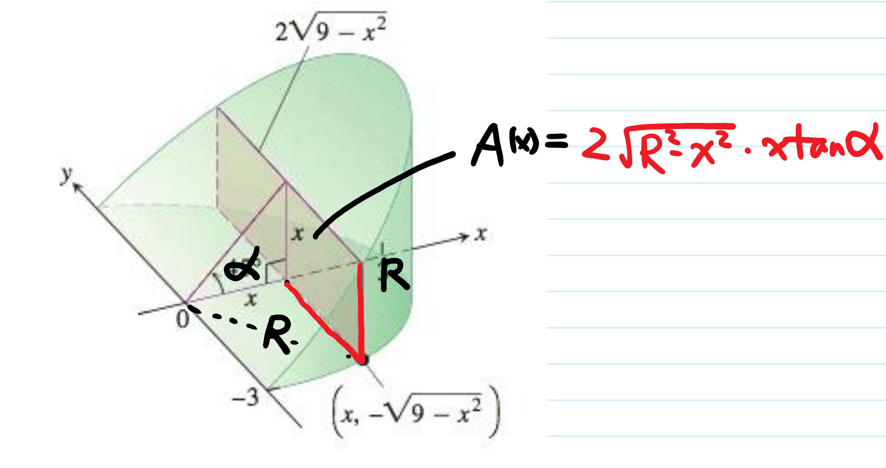
$$V_y = \int_0^4 2\pi x\cdot\sqrt{x}\,dx = 2\pi\int_0^4 x^{\frac{3}{2}}\,dx$$

(2) 平行截面面积为已知的立体体积



$$V = \int_a^b A(x)\,dx$$

例10 一平面经过半径为R的圆柱体的底圆中心，并与底面夹角为α. 计算该平面截圆柱体所得立体的体积（劈锥体）



解法1.（截面为矩形）

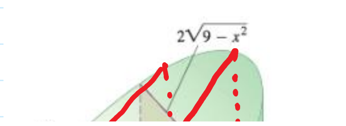
$$V = \int_0^R 2\sqrt{R^2-x^2}\,x\tan\alpha\,dx$$

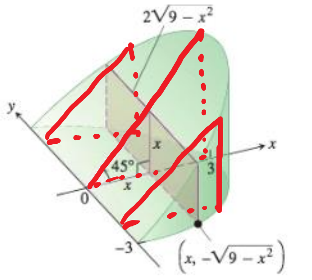
$$= -\tan\alpha\int_0^R \sqrt{R^2-x^2}\,d(R^2-x^2)$$

$$= -\tan\alpha\,\frac{1}{1+\frac{1}{2}}\,(R^2-x^2)^{\frac{3}{2}}\Big|_0^R$$

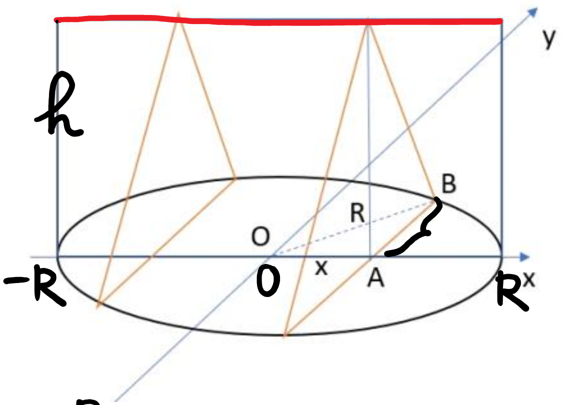
$$= \frac{2}{3}R^3\tan\alpha$$

解法2（截面为直角三角形）





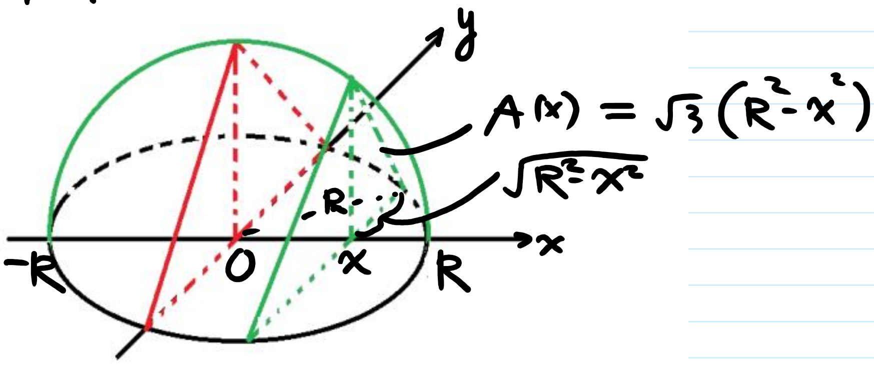
例11. 求以半径为R的圆为底、平行且等于底圆直径的线段为顶、高为h的正劈锥体的体积.



$$V = 2\int_0^R \frac{1}{2}\cdot 2\sqrt{R^2-x^2}\cdot h = 2h\int_0^R \sqrt{R^2-x^2}\,dx$$

$$= 2h\cdot\frac{1}{4}\pi R^2 = \frac{1}{2}\pi R^2 h$$

练习:(4) 底面半径为R. 垂直于直径的截面为等边三角形. 求体积.



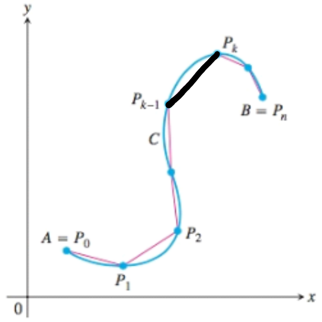
解: $V = 2\int_0^R A(x)\,dx = \cdots$

3. 平面曲线弧长

(1) 定义:

$$M = \lim_{\lambda \to 0} \sum_{i=1}^{n} |P_{i-1}P_i|$$

$$\lambda = \max_{1 \le i \le n} \{ |P_{i-1}P_i| \}$$



(2) 计算: (下限 < 上限)

(i) $L: \begin{cases} x = \varphi(t) \\ y = \psi(t). \end{cases} \quad \alpha \le t \le \beta$ 弧微分

$$s = \left(\int_L ds = \right) \int_{\alpha}^{\beta} \sqrt{\varphi'^2(t) + \psi'^2(t)}\, dt$$

(ii) $L: \quad y = f(x), \quad a \le x \le b.$

$$s = \int_a^b \sqrt{1 + f'^2(x)}\, dx$$

(iii) $L: \quad r = r(\theta), \quad \alpha \le \theta \le \beta$

$$s = \int_{\alpha}^{\beta} \sqrt{r^2(\theta) + r'^2(\theta)}\, d\theta$$

例12. 求星形线 $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ $(a>0)$ 的全长.

解: 参数方程 $\begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases}$

$$s = 4s_1 = 4\int_0^{\frac{\pi}{2}} 3a \sin t \cos t \, dt = 6a.$$

例13. 计算曲线 $y = \int_0^{\frac{x}{n}} n\sqrt{\sin\theta}\, d\theta$ 的弧长 $(0 \le x \le n\pi)$.

$$= f(x)$$

解: $S = \int_0^{n\pi} \sqrt{1+\sin\frac{x}{n}}\, dx$　　　$\frac{x}{n} = t$, $x = nt$

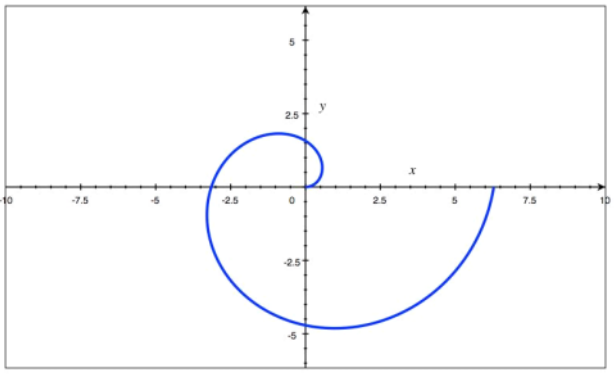
$$= n\int_0^{\pi} \sqrt{1+\sin t}\, dt$$

$$= n\int_0^{\pi} \sqrt{\left(\sin\frac{t}{2} + \cos\frac{t}{2}\right)^2}\, dt$$

$$= n\int_0^{\pi} \left(\sin\frac{t}{2} + \cos\frac{t}{2}\right) dt = 4n$$

练: (5) 证明: 正弦线 $y = a\sin t$ $(0 \le t \le 2\pi)$ 的弧长等于椭圆 $\begin{cases} x = \cos t \\ y = \sqrt{1+a^2}\,\sin t \end{cases}$ $(0 \le t \le 2\pi)$ 的周长.

(6) 求螺线 $r = a\theta$ $(a>0,\ 0 \le \theta \le 2\pi)$ 的弧长.



小结:

1. 面积

$A = \int_a^b f(x)\, dx$　　　$L: y = f(x)$

$= \int_\alpha^\beta y(t)\, x'(t)\, dt$　　　$L: \begin{cases} x = x(t) \\ y = y(t) \end{cases}$ $\alpha \le t \le \beta$

$= \int_{\alpha} y(t)\, x'(t)\, dt$　　$L: \{ y = y(t)$

$= \int_{\alpha}^{\beta} \frac{1}{2} r^2(\theta)\, d\theta.$　　$L: r = r(\theta),\ \alpha \le \theta \le \beta.$

2. 体积：

$$V_x = \int_a^b \pi f^2(x)\, dx \quad y=f(x) \text{ 绕 } x \text{ 轴旋转}$$

$$V_y = \int_c^d \pi \left(R^2(y) - r^2(y)\right) dy \quad \begin{matrix} x=R(y) \\ x=r(y) \end{matrix} \text{ 绕 } y \text{ 轴旋转}$$

$$= \int_a^b 2\pi x f(x)\, dx. \quad y=f(x) \text{ 绕 } y \text{ 轴旋转}$$

3. 弧长.

(i)　$L: \begin{cases} x = \varphi(t) \\ y = \psi(t). \end{cases}$　$\alpha \le t \le \beta$　弧微分

$$S = \left(\int_L ds = \right) \int_{\alpha}^{\beta} \sqrt{\varphi'^2(t) + \psi'^2(t)}\, dt$$

(ii)　$L: y = f(x),\quad a \le x \le b.$

$$S = \int_a^b \sqrt{1 + f'^2(x)}\, dx$$

(iii)　$L: r = r(\theta),\quad \alpha \le \theta \le \beta$

$$S = \int_{\alpha}^{\beta} \sqrt{r^2(\theta) + r'^2(\theta)}\, d\theta$$